

Exp 3: Playfair Cipher

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    char key[20], text[3];
```

```
    char matrix[5][5] = {
```

```
        {'M','O','N','A','R'},
```

```
        {'C','H','Y','B','D'},
```

```
        {'E','F','G','I','K'},
```

```
        {'L','P','Q','S','T'},
```

```
        {'U','V','W','X','Z'}  
    };
```

```
    int i, j, r1, c1, r2, c2;
```

```
    printf("Enter Keyword: ");
```

```
    scanf("%s", key);
```

```
    printf("Enter Plain Text (2 letters): ");
```

```
    scanf("%s", text);
```

```
    for(i = 0; i < 5; i++)
```

```
    {
```

```
        for(j = 0; j < 5; j++)
```

```
        {
```

```
            if(matrix[i][j] == text[0])
```

```
            {
```

```
    r1 = i;  
    c1 = j;  
}
```

```
    if(matrix[i][j] == text[1])  
    {  
        r2 = i;  
        c2 = j;  
    }  
}
```

```
if(r1 == r2)  
{  
    text[0] = matrix[r1][(c1 + 1) % 5];  
    text[1] = matrix[r2][(c2 + 1) % 5];  
}  
else if(c1 == c2)  
{  
    text[0] = matrix[(r1 + 1) % 5][c1];  
    text[1] = matrix[(r2 + 1) % 5][c2];  
}  
else  
{  
    text[0] = matrix[r1][c2];  
    text[1] = matrix[r2][c1];  
}
```

```
printf("Cipher Text: %c%c\n", text[0], text[1]);
```

```
// Find positions of cipher text
```

```
for(i = 0; i < 5; i++)
```

```
{
```

```
    for(j = 0; j < 5; j++)
```

```
    {
```

```
        if(matrix[i][j] == text[0])
```

```
        {
```

```
            r1 = i;
```

```
            c1 = j;
```

```
        }
```

```
        if(matrix[i][j] == text[1])
```

```
        {
```

```
            r2 = i;
```

```
            c2 = j;
```

```
        }
```

```
    }
```

```
}
```

```
if(r1 == r2)
```

```
{
```

```
    text[0] = matrix[r1][(c1 + 4) % 5];
```

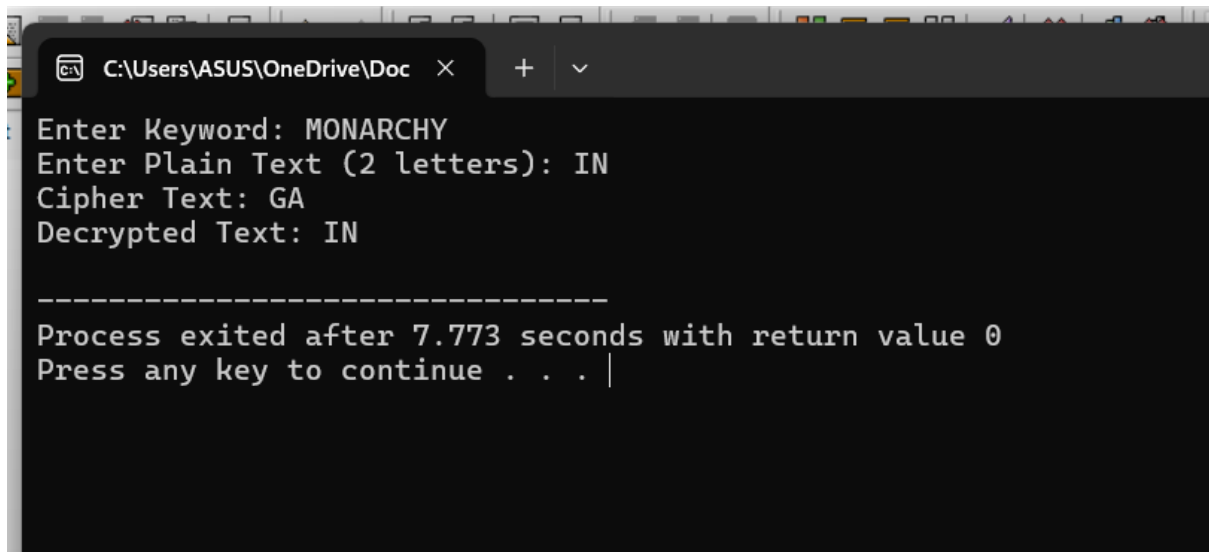
```
    text[1] = matrix[r2][(c2 + 4) % 5];
```

```
}
```

```
else if(c1 == c2)
```

```
{
```

```
    text[0] = matrix[(r1 + 4) % 5][c1];  
    text[1] = matrix[(r2 + 4) % 5][c2];  
}  
else  
{  
    text[0] = matrix[r1][c2];  
    text[1] = matrix[r2][c1];  
}  
  
printf("Decrypted Text: %c%c\n", text[0], text[1]);  
  
return 0;  
}
```



```
Enter Keyword: MONARCHY  
Enter Plain Text (2 letters): IN  
Cipher Text: GA  
Decrypted Text: IN  
  
-----  
Process exited after 7.773 seconds with return value 0  
Press any key to continue . . . |
```